THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE TWO-CHAPTER TEST SET D

Chapters 3 & 4: **Metals and Non-metals, Carbon and its Compounds** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which non-metal is essential for respiration? 1

(a) chlorine **(b)** oxygen **(c)** nitrogen **(d)** sulphur

2. The black coating on silver is: 1

(a) $AgCl$ **(b)** Ag_2S **(c)** Ag_2O **(d)** $AgNO_3$

3. Highly reactive metals are extracted by: 1

(a) reduction with carbon **(b)** roasting only
(c) electrolytic reduction **(d)** heating alone

4. Ionic compounds conduct electricity: 1

(a) never **(b)** in solid state
(c) only as gases **(d)** in molten state and aqueous solution

5. The green coating on copper vessels is due to: 1

(a) copper sulphate **(b)** basic copper carbonate
(c) copper oxide **(d)** copper nitrate

6. The number of carbon atoms in a molecule of propanone is: 1

(a) 3 **(b)** 2 **(c)** 5 **(d)** 4

7. Cyclohexane has the formula: 1

(a) C_6H_{14} **(b)** C_6H_{12} **(c)** C_6H_{10} **(d)** C_6H_6

8. Ethanol on dehydration gives: 1

(a) ethanal **(b)** ethyne **(c)** ethene **(d)** ethanoic acid

9. Soaps are sodium or potassium salts of: 1

(a) long-chain carboxylic acids **(b)** mineral acids
(c) short-chain alcohols **(d)** sulphonic acids

10. The structural isomers possible for C_4H_{10} are: 1

(a) 3 **(b)** 2 **(c)** 1 **(d)** 4

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. Why is the electrical conductivity of ionic compounds zero in the solid state? 2

12. What is the role of a flux? (Name the substance formed.) 2

13. Name the gas evolved when ethanoic acid reacts with $NaHCO_3$ and give its test. 2

14. What is a hydrocarbon? Give one example each of saturated and unsaturated. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. How is aluminium extracted? Write the reactions at the cathode and anode. 3

16. Write the molecular formula and structure of the first member of the homologous series of 3
(i) alcohols
(ii) aldehydes
(iii) ketones.

17. What is meant by the reactivity series? Arrange Na, Cu, Fe, Zn and Ag in decreasing order of reactivity. 3

OR

Give reasons: 3
(i) Metals are used for making bells.
(ii) Electric wires are made of copper.
(iii) Graphite is used for electrodes.

18. Why is graphite used as a lubricant while diamond is used for cutting glass? 3

OR

Explain why soap solution in water forms a colloid and why micelles do not come together to precipitate. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. (i) Describe the electrolytic refining of copper with a labelled set-up. 5
(ii) Write the reactions at the electrodes.

OR

(i) Give four physical properties of metals. 5
(ii) Name two exceptions to these properties.
(iii) Why is the use of metals important in daily life? (one example)

20. (i) Explain covalent bonding with H_2, O_2 and N_2 examples. 5
(ii) Why do covalent compounds have low melting points?

OR

(i) What are soaps and detergents? 5
(ii) Explain micelle formation with a diagram.
(iii) Why do detergents work in hard water?

END OF QUESTION PAPER — **Total: 40 Marks**